For example, we all use GPS for navigation, which suggests an alternative route in real time after receiving information on traffic ahead. Other popular examples include Android Auto, Apple CarPlay and Amazon Alexa available on smartphones to integrate data and improve connectivity.

According to a report from MarketWatch, the global connected passenger car market is expected to reach US$ 75.67 billion by 2023, expanding at a CAGR of eleven per cent, while the global connected truck market is expected to reach US$ 46.84 billion expanding at a CAGR of thirteen per cent in the forecast period from 2018 to 2023. In this article, we focus on connected vehicles (CVs), the key to the next generation of intelligent transportation systems.

Vehicle radar test system (VRTS) from NI (Credit: www.businesswire.com)

Connected and autonomous vehicles

CVs use wireless communication technologies (like short-range radio signals) or devices that allow them to communicate bi-directionally with everything (V2X) inside as well as outside the vehicle including the driver, other vehicles (V2V), pedestrians (V2P) and roadside infrastructure (V2I). Data sharing with other systems and then analysis on the cloud (V2C) enables efficient navigation, security, infotainment, remote diagnostics and payment features, among others.

With AI-enabled services for human-to-vehicle interactions, vehicles become truly smart. One such example is BMW's intelligent personal assistant (IPA), an infotainment system that can be started with the wake-up words 'Hey BMW.' Using ML, the assistant also improves its customised recommendations for the driver.

Unlike a CV where the final decision is made by the driver, a fully autonomous vehicle (AV), popularly known as self-driving vehicle, navigates independently on the road without a human driver and has features like self-parking and auto collision avoidance. But some form of connectivity via devices like sensors, such as light detection and ranging (LIDAR) and radio detection and ranging (RADAR), is necessary to provide real-time information about latest changes on the route and detect operational issues, thereby ensuring reliability.

Most commercial applications still prefer a human operator to handle difficult situations in AVs. Yet, cases like the recent arrest of ex-Waymo driver for deliberately crashing Waymo vehicle into his car and the fatal accident caused by Uber's self-driving car in 2018 show that a lot of work needs to be done in this sector. Many leading automotive (for example, Volkswagen and Toyota) and ICT manufacturers, academia and governments are working in collaboration to make commercial AVs a reality.

Future of smart cities

Intelligent transportation and mobility systems can tackle some of the biggest challenges that make them not a choice for an enhanced consumer experience but a necessity for smart cities. For instance, Toyota Mobility Foundation (TMF) recently entered into an agreement with Malaysia Digital Economy Corp. (MDEC) to launch City Architecture for Tomorrow Challenge (CATCH). The challenge, which began with an event in Kuala Lumpur in February, is aimed to attract innovative solutions from all over the world and will last eight months.

Saving lives. According to 'Pune-Mumbai Expressway Road Accident Study 2018' by JP Research India (JPRI), human errors (not wearing seatbelts, driver fatigue and speeding) and infrastructural flaws were the main reasons for accidents on Pune-Mumbai expressway.

Instead of finding remedies after crashes, smart connectivity improves vehicle safety and efficiency through continuous monitoring of surroundings for situational awareness. It looks out for even those problems that the driver cannot see, like cars in the driver's blind side or change in a traffic light ahead [unlike radar that depends on line-of-sight (LOS) communication and is more expensive].

Due to onboard advanced warning and driving assistance systems (ADASs), drivers have to focus on fewer tasks besides getting enough time and tools to act immediately in dangerous situations. Hence, there is a lower risk of accidents. For instance, in TECO Line Streetcar System, an electric trolley service in Florida, drivers are alerted in emergencies. This is highly useful in cases where continuous driving can lead to lapses in attention.

Low carbon mobility. Transportation is a major contributor to greenhouse gas (GHG) emissions with traffic caused by large vehicle population resulting in a huge amount of fuel wastage. CVs can monitor vehicle type and emissions, and help make eco-friendly decisions like rescheduling trips for congestion reduction and use of public (high occupancy, low emission vehicles) or green (ride-sharing, electric mobility) transportation options. Auto Expo 2020 saw the first-ever electric sports vehicle Marvel X by MG Motor with an electric powertrain, integrated Internet connectivity and autonomous capabilities.